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Mukesh Patel School of Technology, Management & Engineering (Mumbai Campus)

Cyber-Security Department

Project Synopsis

Title of the project:

PrivacyPilot: An AI-Powered DPDP Act Compliance Engine for Automated Code-Aware Privacy Auditing

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Under the Supervision of:

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Table of Contents

1. Title of the project……3
2. Team Members……3
3. Introduction……..3
4. Literature Survey……...…..4
5. Why is this particular topic chosen?.....7
6. Objective and Scope of the Project……..…....8
    6.1 Objectives…….…9
    6.2 Scope of the Project…….…..9
    6.3 Future Work……..……10
7. Hardware and Software Requirements……..…….11
    7.1 Hardware Requirements……...……11
    7.2 Software Requirements……...…….12
        7.2.1 Frontend Technologies……12
        7.2.2 Backend Technologies……...…….12
        7.2.3 AI / NLP Components……..……..13
        7.2.4 Development Environment……....…….13
8. Final Deliverables……..14
9. References……..17

1. Title of the Project: PrivacyPilot - An AI Powered DPDP Act Compliance Engine for Automated Code-Aware Privacy Auditing

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3. Introduction

In today's digital ecosystem, organisations routinely collect and process vast amounts of personal data, including names, email addresses, phone numbers, financial details, government-issued identifiers, health records, and location information. This data powers digital services, personalised experiences, analytics, and business decision-making. However, the increasing volume of personal data collection has also raised significant concerns regarding user privacy, data misuse, unauthorised sharing, and cybersecurity threats.

To address these concerns, the Government of India enacted the Digital Personal Data Protection (DPDP) Act, 2023, establishing the country's first comprehensive legal framework for governing the collection, processing, storage, and transfer of personal data. The Act requires organisations, known as Data Fiduciaries, to collect personal data only with valid consent, process it solely for specified lawful purposes, maintain appropriate security safeguards, ensure transparency, and respect the rights of Data Principals. Failure to comply with these obligations can result in substantial financial penalties of up to ₹250 crore per violation and reputational damage.

Despite these regulatory requirements, many Indian startups and small-to-medium enterprises (SMEs) struggle to evaluate whether their software systems comply with the DPDP Act. Compliance audits are typically performed manually by legal and security experts, making the process expensive, time-consuming, and difficult to scale. Existing privacy governance platforms such as *OneTrust* and *Privado.ai* primarily focus on international regulations like GDPR and are often financially inaccessible for smaller organisations. Moreover, these solutions generally lack automated mapping to the specific provisions of India's DPDP Act.

A significant challenge in privacy compliance lies in the disconnect between an organisation's documented privacy policy and its actual technical implementation. While companies may publicly declare that they collect only limited user information, their source code and third-party integrations may process additional personal data that is neither disclosed nor supported by appropriate user consent. This

inconsistency, referred to as the consent gap creates a compliance risk that exposes organisations to legal, operational, and security consequences.

With the DPDP Act becoming operational, there is an increasing need for an automated solution capable of examining an organisation's source code alongside its privacy documentation. Such a system should identify personal data flows, verify compliance with key statutory obligations, detect inconsistencies between declared and actual data practices, and provide actionable remediation guidance. Artificial Intelligence and Large Language Models (LLMs) present an opportunity to automate these traditionally manual compliance tasks, making privacy auditing faster, more accurate, and accessible.

PrivacyPilot is an AI-powered compliance auditing platform designed specifically for the Digital Personal Data Protection Act, 2023. The system automatically analyses two primary technical sources, GitHub source code repositories and privacy policy documents to construct a view of how personal data flows within an organisation. The platform employs automated scanners to identify Personally Identifiable Information (PII), classifies sensitive data using Natural Language Processing (NLP) techniques, maps identified data flows to three core provisions of the DPDP Act through a rule-based compliance engine, and detects discrepancies between stated privacy policies and actual software behaviour using semantic similarity techniques. An LLM-based reasoning layer then generates detailed compliance reports, risk assessments, and prioritised remediation recommendations in natural language.

PrivacyPilot addresses an important gap in India's privacy compliance ecosystem by providing an affordable, automated, and technically driven solution tailored specifically to the DPDP Act. Unlike conventional compliance tools that primarily analyse legal documents, PrivacyPilot evaluates the actual implementation of software systems, enabling organisations to identify hidden compliance risks before they lead to regulatory action.

By integrating cybersecurity, software engineering, artificial intelligence, natural language processing, and legal compliance into a unified platform, the project demonstrates how modern AI techniques can simplify complex regulatory requirements. The proposed solution not only reduces the cost and effort associated with compliance audits but also assists organisations in building trustworthy, privacy-conscious software systems while promoting responsible data governance.

4. Literature Survey

The increasing importance of data privacy regulations has led to significant research in automated compliance verification, privacy policy analysis, consent management, and AI-assisted legal reasoning. While several studies have proposed intelligent techniques for improving regulatory compliance, most existing solutions are designed for the European General Data Protection Regulation (GDPR) and primarily focus on document analysis rather than evaluating an organisation's actual software implementation. This review examines existing literature across five major research areas relevant to PrivacyPilot and identifies the research gap addressed by the proposed system.

Automated Compliance Verification

Automated compliance verification has emerged as an active research area with the objective of reducing the manual effort involved in interpreting legal regulations. Amaral et al. (2023) proposed an NLP-based framework that automatically verifies Data Processing Agreements against GDPR requirements by extracting legal obligations and comparing them with contractual documents. The study demonstrated that Natural Language Processing can significantly improve compliance assessment while reducing dependence on manual legal review. However, the proposed system focuses solely on legal documents and does not analyse source code or third-party integrations.

Similarly, Okonicha and Sadovykh (2026) extended automated policy evaluation using sentence classification and semantic similarity techniques to identify GDPR compliance gaps within privacy documents. Although the approach improves policy-level compliance analysis, it remains limited to textual documents and does not verify whether the software implementation actually follows the declared privacy practices. These studies establish that automated compliance checking is feasible but reveal the absence of solutions capable of analysing an organisation's technical infrastructure together with its legal documentation.

Privacy Policy Analysis Using Large Language Models

Privacy policies are often lengthy, complex, and difficult for users and organisations to interpret manually. Recent advances in Large Language Models (LLMs) have enabled automated understanding of legal documents. Chen et al. (2025) demonstrated through a comprehensive evaluation that prompt engineering combined with fine-tuning enables LLMs to outperform conventional NLP models across multiple privacy policy datasets, while also noting the limited prior research on applying LLMs to this domain.

Earlier work by Amaral et al. (2021) introduced AI-enabled completeness checking of privacy policies by verifying whether privacy notices satisfy regulatory disclosure requirements. Likewise, Liu et al. (2021) analysed privacy policies against GDPR Article 13 requirements to determine whether organisations adequately disclose their data collection practices. These approaches validate the effectiveness of AI for analysing legal documents but do not compare privacy policy statements with an application's actual software behaviour.

Consent Gap Detection and Law-to-Code Bridging

One of the most significant challenges in privacy compliance is ensuring consistency between what an organisation states in its privacy policy and what its software actually performs. Li et al. (2025) examined inconsistencies between declared data usage statements and actual consent mechanisms, finding that existing approaches primarily analyse cookie banners and isolated consent interfaces while ignoring complete software systems and databases. The authors confirmed that policy-versus-implementation consistency is a major open research problem.

Separately, "Between Law and Code" (Springer ASE, 2026) identified the lack of effective techniques for connecting legal privacy requirements with software implementation, creating a significant research opportunity for automated law-to-code compliance frameworks. A related contribution — the Data Protection by Design Tool (MDPI Sensors, 2022) — translated GDPR regulations into machine-verifiable rules using semantic knowledge graphs, though it remained GDPR-specific and relied on knowledge graph reasoning rather than modern LLM-based techniques. These studies directly motivate the Consent Gap Detector proposed in PrivacyPilot, which compares structured code-derived data practices against policy text using semantic embeddings.

PII Detection and Classification

Accurate identification of Personally Identifiable Information is essential for privacy compliance. Mishra et al. (2025) proposed a hybrid framework combining rule-based NLP, Named Entity Recognition, and Machine Learning for detecting sensitive information within financial documents, achieving a precision of 94.7% and F1-score of 91.1%. The study demonstrates that combining statistical models with handcrafted rules produces superior results. PrivacyPilot builds on this approach by extending an existing PII detection library with India-specific identifiers including Aadhaar numbers, PAN cards, and Indian mobile number formats, rather than building a classifier from scratch, reducing implementation risk while maintaining detection quality.

LLM-Based Compliance Reasoning

Large Language Models have recently become effective tools for interpreting legal documents and generating compliance reports. Zhu et al. (2024) introduced LegiLM, a domain-specific legal language model fine-tuned on GDPR datasets, demonstrating that specialised models significantly improve regulatory reasoning compared to general-purpose LLMs. Chung et al. (2025) further combined structured knowledge graphs with LLM reasoning in GraphCompliance, showing that integrating symbolic rules with language models provides better compliance decisions than either approach alone. Additionally, Hassani et al. (2024) highlighted the growing importance of LLMs in automating legal compliance tasks, emphasising explainability and structured reasoning. These findings support the use of a LangChain-orchestrated LLM in PrivacyPilot to generate human-readable compliance reports, while leveraging a structured DPDP rule engine for reliable legal reasoning.

Research Gap

The reviewed literature establishes that automated compliance checking, privacy policy NLP, and LLM-based legal reasoning are active and productive research areas. However, a critical gap remains: no existing tool performs automated, code-aware compliance checking against India's DPDP Act 2023. Specifically, no prior system: (a) analyses source code to identify personal data processing; (b) maps findings to specific DPDP Act provisions; or (c) automatically detects the gap between a company's privacy policy claims and its actual software implementation for Indian regulatory requirements. PrivacyPilot addresses this gap.

5. Why is this particular topic chosen?

The topic "*PrivacyPilot: An AI-Powered DPDP Act Compliance Engine for Automated Code-Aware Privacy Auditing*" was chosen because data privacy has become one of the most critical concerns in today's digital world. Organisations increasingly collect and process large volumes of personal information, including names, email addresses, phone numbers, financial records, health information, and government-issued identifiers. With the enforcement of the Digital Personal Data Protection (DPDP) Act, 2023, every organisation handling personal data is legally required to ensure that its data processing practices comply with the provisions of the Act.

Despite these legal requirements, most Indian startups and small-to-medium enterprises (SMEs) lack affordable and automated tools to evaluate their compliance. Existing commercial privacy governance platforms primarily focus on international regulations such as the GDPR and are often expensive and difficult for smaller

organisations to adopt. Furthermore, these tools mainly analyse privacy documents and rarely verify whether an organisation's actual software implementation such as source code and third-party SDK integrations matches the privacy commitments made to users. This creates a significant compliance gap and increases the risk of regulatory penalties of up to ₹250 crore per violation under the DPDP Act.

The project was selected to address this practical and emerging problem by developing an AI-powered compliance platform specifically designed for India's DPDP Act. PrivacyPilot combines cybersecurity, artificial intelligence, natural language processing, software engineering, and legal compliance into a focused solution capable of analysing source code and privacy policy documents, detecting Personally Identifiable Information (PII), identifying consent gaps, mapping findings to selected provisions of the DPDP Act, and generating actionable compliance reports.

From an academic perspective, the project explores the integration of Large Language Models (LLMs), semantic similarity techniques, rule-based compliance reasoning, and automated static code analysis to solve a real-world cybersecurity challenge. From an industry perspective, it offers an affordable, scalable, and practical solution that enables organisations to proactively assess and improve their privacy compliance posture. The Consent Gap Detector, which automatically identifies inconsistencies between a privacy policy and the actual behaviour of source code is a novel technical contribution with no direct equivalent in the existing literature.

Therefore, this topic was chosen because it addresses a current national regulatory requirement, fills an identified research gap in automated DPDP compliance assessment, and provides an opportunity to develop an innovative, industry-relevant solution with significant practical and societal impact.

6. Objective and Scope of the Project

The primary objective of PrivacyPilot is to develop an AI-powered compliance auditing platform that assists organisations in evaluating their adherence to the Digital Personal Data Protection (DPDP) Act, 2023 through automated analysis of source code repositories and privacy documentation.

6.1 Objectives

The specific objectives of the project are:

1. To implement a source code scanner capable of analysing GitHub repositories using static analysis techniques, identifying PII-related variable names, function calls, and third-party SDK integrations that process personal data.
2. To develop a privacy policy parser that extracts declared data collection claims and stated purposes from privacy policy documents using NLP techniques.
3. To implement a PII detection module using Microsoft Presidio extended with India-specific identifiers (Aadhaar, PAN, Indian mobile number formats) to classify personal data found in source code.
4. To design a DPDP Rule Engine scoped to three well-defined legal obligations: Consent Requirements (Section 6), Purpose Limitation (Section 8(3)), and Data Minimisation (Section 8(4)), providing a focused, legally defensible, and testable rule set.
5. To develop the Consent Gap Detection module, the primary novel contribution which compares structured data-practice representations derived from source code against policy claims using semantic sentence embeddings (sentence-transformers) and cosine similarity (FAISS) to identify undisclosed data collection or sharing activities.
6. To integrate a LangChain-orchestrated LLM to generate human-readable compliance reports, explain identified violations, provide prioritised remediation steps, and support simple Q&A over generated reports.
7. To design a React-based compliance dashboard displaying compliance scores, violation summaries, risk levels, and downloadable PDF reports.
8. To validate the system using 2–3 synthetic test applications with deliberately planted compliance violations, evaluating precision and recall of the Consent Gap Detector against a known ground truth.

6.2 Scope of the Project

The scope of PrivacyPilot is focused on providing an automated compliance auditing solution specifically designed for organisations operating under the DPDP Act, 2023. In line with the feedback received, the project adopts a focused, deliverable scope for the six-month timeline, centred on the two primary technical sources and a well-defined DPDP rule subset.

The project includes the following functional scope within the primary timeline:

- Source code analysis of GitHub repositories to detect personal data processing, PII variable usage, and third-party SDK integrations.

- Parsing and analysis of privacy policy documents to extract declared data collection practices and stated purposes.
- Identification and classification of PII relevant to Indian privacy regulations using Microsoft Presidio extended with India-specific patterns.
- Mapping of detected data practices to three core DPDP Act provisions: Consent (Section 6), Purpose Limitation (Section 8(3)), and Data Minimisation (Section 8(4)).
- Consent Gap Detection by comparing structured code-derived data flow representations against parsed privacy policy claims using semantic similarity.
- AI-assisted compliance report generation using a LangChain-orchestrated LLM with simple Q&A over findings.
- Interactive web dashboard for reviewing violations, compliance scores, and exporting PDF reports.
- Evaluation using 2–3 synthetic test applications with known, injected compliance violations to measure system accuracy.

The following are explicitly excluded from the primary scope and noted as limitations: live API scanning via OWASP ZAP, database schema inspection, full DPDP Act coverage beyond the three specific provisions, advanced real-time conversational chatbot functionality, and scanning of private organisational codebases. All results and outputs of this system are intended for research and demonstration purposes only and do not constitute legal advice or a substitute for a professional legal audit.

6.3 Future Work

The following capabilities are identified as extensions beyond the primary six-month scope and are intended for future development:

- API/Swagger scanning: lightweight parsing of OpenAPI/Swagger specification files to detect PII fields in API request and response schemas, without live API scanning.
- Live API scanning via OWASP ZAP: dynamic endpoint analysis for runtime PII detection, a significant extension requiring additional DevOps effort.
- Expanded DPDP Rule Engine coverage: extending the rule engine to cover additional DPDP Act obligations beyond the three core provisions, as legal guidance and precedent on the Act mature.

- Full conversational compliance chatbot: a context-aware dialogue system (beyond simple Q&A over reports) providing interactive, session-persistent compliance guidance.

7. Hardware and Software Requirements

7.1 Hardware Requirements

The proposed PrivacyPilot platform is intended to run on a standard development workstation during implementation and testing. Since the application performs static code analysis, Natural Language Processing (NLP), AI-assisted reasoning, and database operations, a moderately powerful system is recommended.

Minimum Hardware Requirements

Component	**Specification**
Processor	Intel Core i5 (10th Generation) / AMD Ryzen 5 or equivalent
RAM	8 GB DDR4
Storage	256 GB SSD
Graphics	Integrated Graphics
Internet	Broadband Internet Connection (Minimum 20 Mbps)

Recommended Hardware Requirements

Component	**Specification**
Processor	Intel Core i7 (12th Generation or above) / AMD Ryzen 7
RAM	16 GB DDR4 or higher
Storage	512 GB SSD or higher
Graphics	NVIDIA GPU (Optional, for faster AI inference)
Internet	High-speed Broadband Connection (50 Mbps or above)

7.2 Software Requirements

The PrivacyPilot platform follows a modern client-server architecture consisting of a React frontend, FastAPI backend, AI/NLP modules, and a database layer. The

selected software components support static code analysis, privacy policy parsing, PII detection, DPDP rule mapping, consent gap detection, and AI-assisted compliance reporting.

Operating System

- Windows 11 (Development)
- Ubuntu 22.04 LTS (Deployment)
- macOS (Supported)

Programming Languages

- Python 3.11+
- JavaScript (ES6)
- SQL

7.2.1 Frontend Technologies

- React.js
- HTML5
- CSS3
- Vite (Build Tool)
- React Query (Data Fetching)
- Recharts (Compliance Score Visualisation)/ React Flow

7.2.2 Backend Technologies

- FastAPI (REST API Framework)
- SQLAlchemy (ORM)
- Pydantic (Data Validation)
- PyGitHub (GitHub Repository Access)
- OpenAPI Specification Validator
- PyMuPDF (PDF Privacy Policy Parsing)
- WeasyPrint (PDF Report Generation)
- PostgreSQL (Database)
- Redis (Caching)
- Celery (Background Task Processing)

7.2.3 AI / NLP Components

- LangChain (LLM Orchestration Framework)
- Large Language Model: Claude API / OpenAI API
- Microsoft Presidio (PII Detection, extended with India-specific patterns)
- spaCy (Supporting NLP Pipeline)
- Sentence Transformers: all-MiniLM-L6-v2 (Semantic Embeddings for Consent Gap Detection)
- FAISS (Vector Similarity Search)

7.2.4 Development Environment

The project is developed using Visual Studio Code as the Integrated Development Environment (IDE). Docker and Docker Compose are used to containerised the backend services, including FastAPI, PostgreSQL, Redis, and supporting containers, ensuring consistent development and deployment environments. Git and GitHub are used for source code management and collaborative development. Postman is employed for testing RESTful APIs, while PyTest and FastAPI TestClient are used for automated unit and integration testing.

Category	**Technology Used**
Operating System	Windows 11 / Ubuntu 22.04
Frontend	React.js, HTML5, CSS3, Vite
Backend	Python 3.11, FastAPI, SQLAlchemy
Database	PostgreSQL
AI Framework	LangChain
PII Detection	Microsoft Presidio (+ India-specific extensions)
NLP Library	spaCy
Vector Search	FAISS, Sentence Transformers
Code Scanner	PyGitHub, AST (Python built-in)
PDF Processing	PyMuPDF, WeasyPrint
Caching	Redis
Containerisation	Docker, Docker Compose
Version Control	Git, GitHub

Testing	PyTest, Postman, FastAPI TestClient
IDE	Visual Studio Code

8. Final Deliverables

The final deliverable of this project is a fully functional prototype of PrivacyPilot, an AI-powered compliance auditing platform designed to help organisations assess their compliance with the Digital Personal Data Protection (DPDP) Act, 2023. The system will integrate automated source code analysis, privacy policy evaluation, rule-based compliance mapping for three core DPDP provisions, and AI-assisted report generation into a single web-based platform.

The completed project will include the following deliverables:

1. Source Code Scanner

A static analysis engine that clones a GitHub repository and scans Python and JavaScript files to identify PII-related variable names, field declarations, and third-party SDK calls that process personal data.

2. Privacy Policy Parser

An NLP module that downloads a privacy policy (URL or PDF) and extracts structured claims about what data is collected, for what purpose, and with which third parties, producing a machine-comparable representation of stated data practices.

3. PII Detection and Classification Module

A module using Microsoft Presidio extended with India-specific PII patterns (Aadhaar, PAN, Indian mobile numbers, financial identifiers) to classify personal data found in source code by DPDP PII category and sensitivity level.

4. Consent Gap Detection Module

The primary novel contribution of this project. An intermediate representation layer converts raw code findings into structured "data practice" objects and raw policy text into structured "policy claim" objects. Semantic sentence embeddings (sentence-transformers: all-MiniLM-L6-v2) are generated for both, and FAISS cosine similarity matching identifies source code data practices with no semantically corresponding policy claim, indicating undisclosed data collection, processing, or sharing activities. This two-stage (structure → embed → compare) pipeline directly addresses the modality gap between natural-language policy text and structured code signals.

5. AI-Powered Compliance Report Generator

A LangChain-orchestrated LLM module that receives all structured findings (rule engine results + consent gap detections) and generates a human-readable compliance report with section-cited DPDP violations, risk assessments, remediation recommendations, and an overall compliance score (0–100). Includes a simple Q&A interface: users select a violation or report section and ask a predefined set of question types; the system retrieves relevant findings via RAG and generates a contextual response.

6. Web-Based Compliance Dashboard

A React-based dashboard allowing organisations to connect their GitHub repository and upload a privacy policy document. Displays a compliance score gauge, a DPDP section-wise violation list with evidence and severity, and a downloadable PDF compliance report. A simple Q&A interface is available within the dashboard.

7. Evaluation and Validation

Validation is conducted using 2–3 purpose-built synthetic test applications with deliberately injected DPDP compliance violations of known type and location. Ground truth is established manually before scanning. Precision, recall, and F1-score of the Consent Gap Detector are measured against this ground truth. This controlled evaluation approach provides a clear, reproducible benchmark without requiring access to private organisational codebases.

8. Project Documentation

Complete software documentation including system architecture diagrams, database schema, API documentation, installation guide, user manual, testing reports, and the final capstone project report.

The completed PrivacyPilot prototype will provide organisations with an automated, focused, and affordable solution for evaluating key aspects of their privacy compliance under the DPDP Act, reducing manual audit effort while improving transparency and regulatory preparedness.

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2026–2027

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